

A machine-certified exhaustion of the first exact identity basis for hypothetical Ramsey(5, 5, 45) graphs, and the measured obstruction that closes it

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Abstract

We report the terminal outcome of an exact, fully machine-verified linear programming campaign over the finite state relaxation of a hypothetical Ramsey(5, 5, 45) graph. The relaxation carries one interval row per subgraph-counting identity: the McKay–Radziszowski (MR97) rows at $m = 2, 3, 4$ and Engström’s non-separable degree-4 neighbourhood/anti-neighbourhood identity, the only identity of the campaign’s recorded first exact basis that had never been used inside any computational certificate. Adding the Engström row produces *no* accepted structural consequence: all three instantiable routes are rejected by exact rational primal witnesses, so the recorded basis is exhausted at $n = 45$. We claim nothing about $R(5, 5)$.

The contribution is not the negative disposition but its mechanism, which we measure exactly. First, the new row is genuinely new geometrically: the rank of the state-endpoint matrix rises from 6 to 8 when the Engström endpoints are adjoined, refuting the natural hypothesis that projection onto the $n = 45$ domain collapses its coordinates. Second, it nonetheless has *zero* cut strength: the exact dual and primal optima of all three routes are bit-identical with and without it (360, 14310/349, 45), and every optimal certificate carries $\eta = \theta = 0$. Third, we localise the loss: it is the per-coordinate interval abstraction, not the identity or the projection. The median per-state Engström window is wider than the median magnitude of the row itself (ratio 1.207), its aggregate constraints are strictly slack at every optimal witness, and on a complete catalog class the discarded joint structure is a $\rho = 0.9464$ -correlated point cloud replaced by its bounding box.

Fourth, we show that fixing any of this would not help. Although the envelope certification of the 77 catalog-missing stratum classes is $111\times$ looser than catalog truth and carries 98.6% of all window width, collapsing every such window to a *point* still leaves the three routes at 360, 39.25, 38.36 against acceptance edges 315, 1, 1: a factor- ≈ 39 gap survives perfect data. The obstruction is not window quality but the absence of vertex-to-vertex coupling.

Fifth, we therefore build the level-2 pair-coupled lift and close it too. With adjacent and non-adjacent degree-pair variables, the per-degree-class neighbour-degree identity and three exact codegree bounds, the optima move to 360, 41, 45: two routes unchanged, one improved by $1/349$. The explanation is exact — the campaign’s $m = 2$ row *is* the degree-squared identity $\sum_v \deg(v)^2 = 45e + \sum_v e(G[N(v)]) - \sum_v e(G[D(v)])$ in disguise, so every handshake-style repair was already present. Levels 1 and 2 of the first exact basis at $n = 45$ are thus both closed. We also record, with its regression test, a near-miss row that is independent of the basis and appears to cut hard but is unsound: exact certificates prove optimality for the programme written, not soundness of the programme.

*Computational campaign; all artifacts, sources and verification transcripts accompany this note.

1 Setting

Write $R(5, 5) \in \{43, \dots, 46\}$ (Exoo; Angeltveit–McKay [3]). At $n = 45$ the only published structural restriction on a hypothetical Ramsey(5, 5, 45) graph is the degree window $d_v \in \{20, \dots, 24\}$. Our campaign works inside a finite relaxation of that setting, built from complete published $R(4, 5)$ censuses.

Definition 1 (State relaxation). For a hypothetical Ramsey(5, 5, 45) graph G and a vertex v put $d = d_v$, $m = 44 - d$, let $X = G_v^+$ be the neighbourhood, $Y = G_v^-$ the anti-neighbourhood, and $Z = \bar{Y}$. Both X and Z are $R(4, 5)$ graphs, so $e(X) = E_{45}(d) - a$ and $e(Z) = E_{45}(m) - b$ for edge deficiencies $a, b \geq 0$, where E_{45} is the published maximum edge count. The *state* of v is (d, a, b) ; there are exactly 3,215 feasible states. Each state carries exact integer intervals for four rows: the $m = 2$ excess balance (a single integer), the $m = 3$ row g , the $m = 4$ row h , and the Engström mixed row F of Section 2.

Because every vertex occupies exactly one state, G induces nonnegative rational weights w_s with

$$\sum_s w_s = 45, \quad \sum_s w_s \text{bal}(s) = 0, \quad \sum_s w_s r_{lo}(s) \leq 0 \leq \sum_s w_s r_{hi}(s) \quad (r \in \{g, h, F\}), \quad (1)$$

since each identity sums to zero over the 45 vertices while each vertex row lies in its own state window. All statements below concern this relaxation only.

2 The Engström row

Engström [2] proved the MR97 subgraph-counting conjecture, yielding for every graph G on n vertices the universal identity $\sum_v [p_1(G_v^+) + p_2(G_v^-) + p_3(G_v^+, G_v^-)] = 0$ with

$$\begin{aligned} p_1(X) &= n(n-3)s(K_1, X) - (n^2 + 2n - 6)s(K_1, X)^2 + 3n s(K_1, X)^3 - 2s(K_1, X)^4 \\ &\quad + 2(n^2 + n - 8)s(K_2, X) - 12s(K_2, X)^2 - 12(n-1)s(K_1, X)s(K_2, X) + 12s(K_1, X)^2s(K_2, X) \\ &\quad + 72s(C_4, X) + 12(n-2)s(K_3, X) + 24s(K_{1,3}, X) + 24s(P_4, X) + 24s(T_{3,1}, X) \\ &\quad + 12(n+2)s(P_3, X) - 24s(K_1, X)s(P_3, X) + 32s(T_{3,2}, X), \\ p_2(Y) &= 4s(K_2, Y)^2 - 12s(K_{1,3}, Y) - 8s(C_4, Y) - 8s(T_{3,1}, Y) - 24s(T_{3,2}, Y) + 2(n-8)s(P_3, Y), \\ p_3(X, Y) &= 4s(K_1, X)s(P_3, Y) - 2(n-2)s(K_1, X)s(K_2, Y) + 4s(K_1, X)^2s(K_2, Y), \end{aligned}$$

where $s(J, H)$ counts induced copies, $T_{3,1}$ is the paw and $T_{3,2}$ the diamond. Two features matter here. It is *non-separable* (the p_3 cross terms), hence outside the MR97 Lemma 2.3 classification of separable identities of degree ≤ 6 ; and its coordinate support $(C_4, K_{1,3}, P_3, P_4$ on both sides, plus three mixed products) is disjoint from the frozen basis support $\{d, e, t, \text{paw}, \text{diamond}, i_4\}$. It requires no K_4 -free hypothesis. To our knowledge it had never been used inside a computational certificate for any Ramsey claim.

Specialising to $n = 45$ and eliminating the Y -side by the complement map $s(P_3, \bar{Z}) = \text{pc}_3(Z)$, $s(C_4, \bar{Z}) = s(2K_2, Z)$, $s(K_{1,3}, \bar{Z}) = s(K_3 + K_1, Z)$, $s(T_{3,1}, \bar{Z}) = s(P_3 + K_1, Z)$, $s(T_{3,2}, \bar{Z}) = s(K_2 + 2K_1, Z)$, $s(P_4, \bar{Z}) = s(P_4, Z)$ gives the per-vertex row used throughout, with $e_y = \binom{44-d}{2} - e_z$:

$$\begin{aligned} F_v &= 1890d - 2109d^2 + 135d^3 - 2d^4 + 4124e_x - 12e_x^2 - 528de_x + 12d^2e_x + 4e_y^2 - 86de_y + 4d^2e_y \\ &\quad + 516t(x) + 72c_4(x) + 24\text{claw}(x) + 24p_4(x) + 24\text{paw}(x) + (564 - 24d)p_3(x) + 32\text{diamond}(x) \\ &\quad + (74 + 4d)\text{pc}_3(z) - 12k_3k_1(z) - 8(2K_2)(z) - 8(P_3 + K_1)(z) - 24(K_2 + 2K_1)(z). \end{aligned} \quad (2)$$

Kernel validation. The eleven raw induced 4-vertex counters, the complement map and (2) are verified by exhaustive comparison against naive 4-subset enumeration on all 33,867 labelled graphs with $n \leq 6$, spot checks at $n \in \{7, 8\}$, residual zero on all 328 published Ramsey(5, 5, 42) representatives *and* their complements (656 checks), and replay of the $n = 49$ type-combination values $\{0, 144, 288, 432\}$ with a unique zero corner on the frozen order-24 census. Six plausible closed forms are provably wrong and are pinned as detected mutations with named witnesses (e.g. $\text{claw} = \sum_v \binom{d_v}{3} - \text{paw}$ fails on the 4-vertex diamond; $s(C_4, \bar{Z}) = s(C_4, Z)$ fails on $2K_2$; dropping $+c_{uv}$ from pc_3 fails on K_3).

Interval endpoints. Per-class exact motif windows come from a single streaming pass over 8,500,211 graphs (the 36 validated edge-extremal $R(4, 5)$ catalogs at orders 17–23 plus the complete order-24 census), recording exact histograms of fifteen aggregates and asserting K_4 -freeness on every graph. Classes absent from the catalogs receive sound outer windows from a certified envelope LP over the R1–R5 relation system (raw 4-motif coordinates) or from degree-histogram enumeration (the 3-motif-derived aggregates). Every observed catalog value is verified to lie inside its outer window. Since (2) splits additively over the two strata, interval endpoints are obtained by sign-aware affine images of the per-coordinate windows.

3 Certificates, witnesses, and the terminal result

Definition 2 (Eight-coefficient certificate). A tuple $(\alpha, \beta, \gamma, \delta, \varepsilon, \zeta, \eta, \theta)$ with $\gamma, \delta, \varepsilon, \zeta, \eta, \theta \geq 0$ such that for every state s

$$\text{obj}(s) \leq \alpha + \beta \text{bal}(s) + \gamma g_{lo}(s) - \delta g_{hi}(s) + \varepsilon h_{lo}(s) - \zeta h_{hi}(s) + \eta f_{lo}(s) - \theta f_{hi}(s).$$

Weighting by any w admissible for (1) gives $\sum_v \text{obj} \leq 45\alpha$; we additionally require α to be the least constant its own coefficients admit, so certificates are canonical.

Three routes are frozen in advance, with acceptance edges fixed before any run: `total_deficiency` (accepted at $45\alpha \leq 315$), `degree20_count` and `deficiency_ge8_count` (accepted at $45\alpha < 1$, which forces a nonnegative integer count to vanish). A fourth route, `required_local_family`, is not instantiable in this repository (no hash-pinned complete gluing-cover manifest exists) and is recorded with no evidence at all. A route is refuted when an exactly rational admissible w attains a value beyond the same edge.

Theorem 3 (Terminal result). *Over the relaxation of Definition 1 with all four rows, every instantiable route is refuted by an exact rational primal witness:*

$$\text{total_deficiency} : 360, \quad \text{degree20_count} : \frac{14310}{349}, \quad \text{deficiency_ge8_count} : 45,$$

each equal to the exactly verified least upper bound 45α of the same route. Consequently no certificate over $\{\text{bal}, g, h, F\}$ can accept any frozen route, and the campaign’s recorded first exact basis (MR97 $m = 2, 3, 4$ plus the Engström identity) is exhausted at $n = 45$.

Verification. Coefficients are proposed by floating-point LP, then discarded: each is rationalised on a frozen denominator ladder, α is re-derived exactly as $\max_s(\cdot)$, and the affine inequality is re-checked at all 3,215 states in exact rational arithmetic. Primal supports are proposed numerically, then re-solved exactly by **Fraction** Gaussian elimination over every active/inactive choice of the six interval inequalities; each witness is re-verified against (1) including the two new aggregate

conditions. Route statuses and the terminal disposition are re-derived from verified evidence, never read from storage. An independent verifier that imports none of the producers re-hashes all 64 recorded inputs, re-derives the 8,500,211 catalog counts across 53 classes, recomputes all 3,215 F intervals from the artifact’s own windows using its own affine arithmetic, and re-derives every bound, witness value, status and the disposition. A twelve-mutation corruption battery (endpoints, padded and lowered α , witness weights, trust roots, histogram counts, replay fields, route status, deleted state, schema version, swapped window pair, input digest) is rejected item-by-item while the untouched artifact passes. $\square$

Remark 4. Theorem 3 is a statement about a relaxation, not about $R(5, 5)$. It refutes routes; it proves no bound. In particular nothing here supports or contradicts $R(5, 5) \leq 45$.

4 The measured obstruction

The negative outcome had two candidate explanations: the new row might be redundant after projection (“coordinate collapse”), or it might be informative but wasted by the relaxation. We separate them exactly.

Proposition 5 (The row is geometrically new). *Over the 3,215 states,*

$$\text{rank} [\mathbf{1}, \text{bal}, g_{lo}, g_{hi}, h_{lo}, h_{hi}] = 6, \quad \text{rank} [\cdots, f_{lo}] = 7, \quad \text{rank} [\cdots, f_{lo}, f_{hi}] = 8.$$

Hence neither Engström endpoint is an affine combination of the frozen endpoints: the collapse hypothesis is false.

Proposition 6 (Yet it has zero cut strength). *For each frozen route the exact optimum of the dual programme and of the primal programme is unchanged when the two F columns, respectively the two F inequalities, are deleted:*

<i>route</i>	<i>dual (m_4 basis)</i>	<i>dual (mixed)</i>	<i>primal (m_4)</i>	<i>primal (mixed)</i>
<i>total_deficiency</i>	360	360	360	360
<i>degree20_count</i>	14310/349	14310/349	14310/349	14310/349
<i>deficiency_ge8_count</i>	45	45	45	45

*Every optimal certificate carries $\eta = \theta = 0$; e.g. the *degree20_count* optimum is $(\alpha, \beta, \delta) = (318/349, 6/349, 1/1047)$ with all other coefficients zero.*

Proposition 7 (Mechanism: the interval abstraction). *The two new aggregate constraints are strictly slack at every optimal witness, with*

$$\sum_s w_s f_{lo}(s) = -\frac{18522745}{2}, \quad -\frac{2245549320}{349}, \quad -11622240$$

for the three routes respectively (and correspondingly large positive $\sum_s w_s f_{hi}$). The slack is structural: per-state window widths are

<i>row</i>	<i>min</i>	<i>median</i>	<i>mean</i>	<i>max</i>
<i>g</i>	0	930	986	1,458
<i>h</i>	0	41,128	41,808	72,078
<i>F</i>	864	802,396	764,953	1,318,306

so the median F window (802,396) exceeds the median $|F|$ magnitude (664,870). The row’s information is destroyed by per-coordinate intervalisation, not by the identity and not by the projection.

Proposition 8 (The discarded structure is large and measurable). *On the complete catalog class (22,114) (133 graphs), the x -side contributions to the h and F rows satisfy $h_x \in [1686, 2056]$, $F_x \in [234048, 239156]$ and*

$$\rho(h_x, F_x) = 0.9464.$$

The box relaxation replaces this near-degenerate cloud by its 370×5108 bounding box.

Propositions 5–8 together give a sharper negative than a bare disposition: the frozen basis is exhausted *as parameterised by independent per-row intervals*, and the exhaustion is caused by a quantified loss of joint information rather than by exhaustion of the identities.

5 Window quality is not the bottleneck

Propositions 7–8 suggest tightening the relaxation: each row splits additively over the two strata, $r = r_{\text{exact}}(d, a, b) + r_x(x) + r_z(z)$ for $r \in \{g, h, F\}$, so the attainable set of the *triple* (g, h, F) is a Minkowski sum of two finite point clouds and the present relaxation replaces it by a bounding box. Three such repairs suggest themselves: convex hulls over the catalog clouds; structural extremal caps on the catalog-missing classes (e.g. Zykov’s theorem gives $i_4(z) \leq 900$ at order 22 against the envelope’s 1285); and exact joint projection of the envelope polytope. We measured all three, and all three are dead ends.

Proposition 9 (Where the width lives). *Summed over all 3,215 states, the h and F window width contributed by catalog-certified stratum classes is 35,649,683 (1.4%); by the 77 envelope-LP classes, 2,558,092,854 (98.6%). Hence the catalog-side correlation of Proposition 8 governs only 1.4% of the width, and hull-tightening over catalog clouds cannot matter. Separately, exact joint projection of the envelope polytope onto the (h_z, F_z) plane for the class (22,107) tightens the area only by a factor 1.5.*

Proposition 10 (The envelope is $111\times$ loose, and it still does not matter). *On the five catalog-present extremal classes the envelope LP is $111\times$ looser than the truth in aggregate width (92,888 against 836; e.g. i_4 at (22,114) is truly [67,99] against the LP’s [0,1285]). Nevertheless, recomputing every h and F endpoint from the artifact’s own window table while shrinking all envelope-class windows toward their midpoints by a factor k (holding g at its frozen values; at $k = 1$ the recomputation reproduces all 3,215 recorded intervals exactly) gives*

k	<i>total_deficiency</i> (edge ≤ 315)	<i>degree20_count</i> (edge < 1)	<i>deficiency_ge8_count</i> (edge < 1)
1	360	41.003	45.000
0.1	360	39.352	38.632
0.01	360	39.260	38.387
0	360	39.251	38.363

with primal = dual throughout; at $k = 0$ the primal still places 39.251 of the 45 vertex-weight on degree-20 states.

Even a *perfect* motif certification — every catalog-missing class pinned to a point — therefore leaves a factor- ≈ 39 gap on the count routes. No data-quality campaign can close it. (The rows $k < 1$

are a sensitivity analysis, not a theorem: the shrunken windows are centred at current midpoints and are not sound certifications. The rigorous row is $k = 1$; the robustness is nonetheless decisive, three orders of magnitude of width reduction buying 1.75 units on a gap of 38.)

What the obstruction is. The relaxation imposes exactly five aggregate conditions on 3,215 nonnegative weights, so an optimal vertex has support at most five, and five linear conditions cannot separate the degree-20 states: the programme is free to place ≈ 39 –41 of the 45 weight on them. The parameterisation is not at fault; the *absence of vertex-to-vertex coupling* is. The relaxation never records that neighbourhoods of adjacent vertices intersect. A cheap candidate repair also fails: combining $\sum_s w_s d_s = 2e$ with the averaged per-vertex inequality $e \geq d_v + e(N_v) + e(D_v)$ is slack by ≈ 200 at every state, since $e \approx 495$ while the right-hand side is ≈ 250 .

More constraints, not better data. What is missing is vertex-to-vertex coupling: a level-2 Sherali–Adams lift carrying pair variables with marginal consistency to the present w_s . We built it, and it is closed.

6 The level-2 pair lift is closed

The lift adjoins, for each integer edge count e , ordered degree-pair variables $M[d, d']$ (adjacent) and $N[d, d']$ (non-adjacent), symmetric and nonnegative, with marginals $\sum_{d'} M[d, d'] = d n_d$ and $\sum_{d'} N[d, d'] = (44 - d)n_d$ where $n_d = \sum_{d_s=d} w_s$; the per-degree-class neighbour-degree identity (3) below; the per-state window $20d_s \leq e + \varphi(s) \leq 24d_s$; and three codegree facts, writing $\lambda(u, v) = |N(u) \cap N(v)|$. For $u \sim v$, $N(u) \cap N(v)$ is K_3 -free with independence number ≤ 4 , so $\lambda \leq R(3, 5) - 1 = 13$. For $u \not\sim v$, $D(u) \cap D(v)$ has independence number ≤ 2 and is K_5 -free, so $|D(u) \cap D(v)| = 43 - d_u - d_v + \lambda \leq R(5, 3) - 1 = 13$, giving the *upper* bound $\lambda \leq d_u + d_v - 30$; and $\lambda \geq \max(0, d_u + d_v - 43)$ since $N(u) \cup N(v)$ omits u and v . These tie to the vertex layer through $\sum_v e(G[N(v)]) = 3T$ and $\sum_{u \not\sim v} \lambda = \sum_v \binom{d_v}{2} - 3T$.

Proposition 11 (The pair lift does not open the routes). *Maximised over every integer e for which the lifted programme is feasible, which is exactly $e \in [454, 536]$, the exact optima are 360, 41 and 45 against frozen 360, 14310/349 and 45. Two routes are unchanged; the third improves by 1/349. No route reaches its acceptance edge.*

The reason is a collapse we can state exactly. For any finite simple graph, pure double counting gives $\sum_{u \in N(v)} \deg(u) = d_v + 2e(G[N(v)]) + e_{xz}(v)$ and $e = d_v + e(G[N(v)]) + e(G[D(v)]) + e_{xz}(v)$, hence for every vertex

$$\sum_{u \in N(v)} \deg(u) = e + e(G[N(v)]) - e(G[D(v)]), \quad (3)$$

and, summing and using $\sum_v \sum_{u \in N(v)} \deg(u) = \sum_u \deg(u)^2$,

$$\sum_v \deg(v)^2 = 45e + \sum_v e(G[N(v)]) - \sum_v e(G[D(v)]). \quad (4)$$

We verified (3) and (4) on all 2,131,018 labelled graphs up to $n = 7$ and on random graphs at $n = 12, 20, 45$.

Proposition 12 (The $m = 2$ row is the degree-squared identity). *Adjoining (4) to (1) gains nothing, because it is already present. In this repository the z stratum is the complement of the anti-neighbourhood, so $e_z = E_{45}[m] - b$ equals $e(\bar{G}[D(v)])$ and the interior count is $e_y = \binom{m}{2} - e_z$; with that substitution $2d^2 - 45d - 2e_x + 2e_y = \text{bal}(s)$ identically on all 3,215 states, and the exact rank of $\{1, \text{bal}, g_{\text{lo}}, g_{\text{hi}}, h_{\text{lo}}, h_{\text{hi}}, f_{\text{lo}}, f_{\text{hi}}\}$ is 8 both before and after adjunction.*

Proposition 12 retires the whole family of handshake-, assortativity- and degree-correlation repairs at the aggregate level, including the one suggested by §5. Only the per-degree-class form of (3) is strictly finer than its aggregate, and Proposition 11 prices that refinement at $1/349$.

A recorded trap. Substituting the recorded e_z directly for $e(G[D(v)])$ in (4) yields $r(s) = 2d^2 - 45d - 2(E_{45}[d] - E_{45}[m]) + 2a - 2b$, which is independent of the frozen basis (rank $8 \rightarrow 9$) and appears to cut every route hard — to 4905/14, 225/11, 4905/112, and under integrality of the degree distribution to 349, 20, 43 — with matching exact dual certificates and exact primal witnesses throughout. All of it is invalid: r misreads the stratum convention. We record it because exact certificates and matching witnesses establish optimality *for the programme written down*; they carry no information about whether that programme is sound. Reading the non-adjacent bound $\lambda \leq d + d' - 30$ as a lower bound is a second such trap, and makes every e infeasible, i.e. manufactures a refutation of $R(5, 5) = 45$. Both are pinned by regression tests.

Consequence. Levels 1 and 2 of this basis are now both closed, and the factor- ≈ 39 margin of Proposition 10 shows data quality cannot substitute for a non-local argument. The recorded residual tiers —VeriPB-certified gluing search, then $\text{srg}(45, 22, 10, 11)$ complementary SAT—are the only remaining routes.

7 Reproducibility

All numbers above are produced by the accompanying sources and artifact. The canonical artifact `engstrom_identity.json` (schema 3) records the provenance of every input by SHA-256, the fifteen per-class exact histograms, the per-class catalog and outer windows with their derivation method, all 3,215 states with interval and motif provenance, the four search records with exact certificates and witnesses, and the disposition. Commands:

```
python -m unittest discover -s r55/tests -p 'test_*mixed*.py' -v
python r55/src/mixed_deficiency_cone.py --output r55/data/engstrom_identity.json
python r55/src/search_mixed_cuts.py      --analysis r55/data/engstrom_identity.json
python r55/src/verify_mixed_independent.py      r55/data/engstrom_identity.json
```

Trust boundary and remaining gap. Catalog completeness is assumed from the published censuses and is named, not re-proved. Floating-point LP output is used only to propose; every accepted number is exact. The disposition is producer-verified, independently re-verified at the search and F -interval layer, and cross-anchored to the frozen $m = 4$ artifact. Not yet redone independently: a second motif kernel re-sweeping the 8.5M graphs, and recomputation of the g, h endpoints from R1–R5 rather than by equality against the frozen $m = 4$ projection.

References

- [1] B. D. McKay and S. P. Radziszowski, *Subgraph counting identities and Ramsey numbers*, J. Combin. Theory Ser. B **69** (1997), 193–209. <https://users.cecs.anu.edu.au/~bdm/papers/r55.pdf>
- [2] A. Engström, *A proof of the McKay–Radziszowski subgraph counting conjecture*, arXiv:1002.4304 (2010). <https://arxiv.org/abs/1002.4304>
- [3] V. Angelteit and B. D. McKay, $R(5, 5) \leq 46$, arXiv:2409.15709 (2024). <https://arxiv.org/abs/2409.15709>
- [4] V. Angelteit and B. D. McKay, $R(5, 5) \leq 48$, arXiv:1703.08768 (2017). <https://arxiv.org/abs/1703.08768>